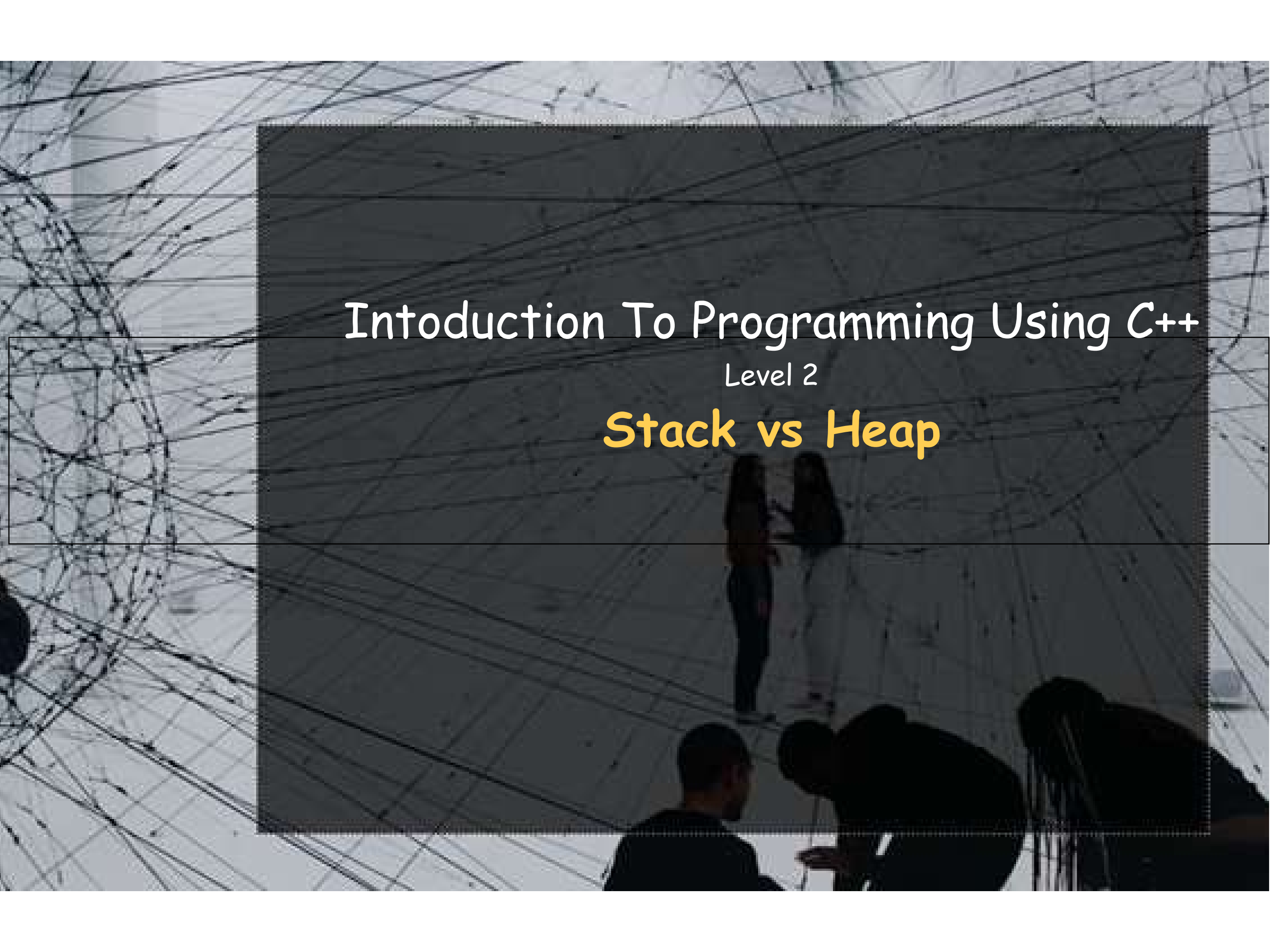


Intoduction To Programming Using C++

Level 2

Stack vs Heap



Introduction To Programming Using C++

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Stack vs Heap

🔍 Definition

🔍 Definition

▣ Stack and Heap

▤ Stack and Heap are two important areas of memory used during program execution.

▤ Each one works differently and is used for different purposes.

Explanation

Memory (RAM)

Source Code / Instructions

-  Stores the program instructions (compiled code).
-  Usually takes a relatively small amount of memory.

Static & Global Variables

-  Contains global and static variables.
-  These variables generally remain available for the lifetime of the program.

Stack

-  Used for local variables and function call information.
-  Works using LIFO (Last In, First Out).
-  Very fast to access.
-  Its available size is limited compared with the Heap.

Heap

-  Used for dynamic memory allocation.
-  In C++, dynamic memory can be allocated using `new` and released using `delete`.
-  It can provide a larger and more flexible memory area than the Stack.
-  Dynamic allocation and deallocation generally have more overhead than Stack allocation.

Note

- ◆ In modern C++, the distinction between Stack and Heap is useful for understanding memory management, but the exact implementation and memory layout depend on the compiler, operating system, and platform.
- ◆ Modern C++ often uses RAII, containers such as `std::vector`, and smart pointers such as `std::unique_ptr` instead of manually managing memory with raw `new` and `delete`.

Comparison

Stack

- ✓ Mainly used for local variables and function call information.
- ✓ Follows LIFO.
- ✓ Very fast allocation and access.
- ✓ Limited in size.
- ✓ Memory lifetime is generally managed automatically according to scope and function calls.
- ✓ Suitable for temporary data.

Heap

- ✓ Used for dynamic memory allocation.
- ✓ Memory can be allocated during run-time.
- ✓ More flexible for objects whose lifetime or size needs to be determined dynamically.
- ✓ Allocation can be slower and have more overhead than Stack allocation.
- ✓ With raw ``new` / `delete``, the programmer is responsible for releasing the memory.
- ✓ Can be used for dynamically allocated data structures.

✦ Summary

- ✓ Stack is fast and automatically managed, but its available size is limited.
- ✓ Heap provides flexible dynamic memory allocation.
- ✓ Stack is commonly used for local variables and function call information.
- ✓ Heap is commonly used for dynamically allocated objects and arrays.
- ✓ ``new`` allocates memory dynamically on the Heap.
- ✓ ``delete`` releases memory allocated with ``new``.
- ✓ Incorrect Heap management can cause memory leaks.
- ✓ Excessive Stack usage, such as very deep recursion, can cause Stack Overflow.

Thank you 😊